YAKEEN

NEET 2.0 2027

Physics | Lecture - 03

Vector

By- Manish Raj (MR Sir)



Todays Goal

Type of vector * Part 2

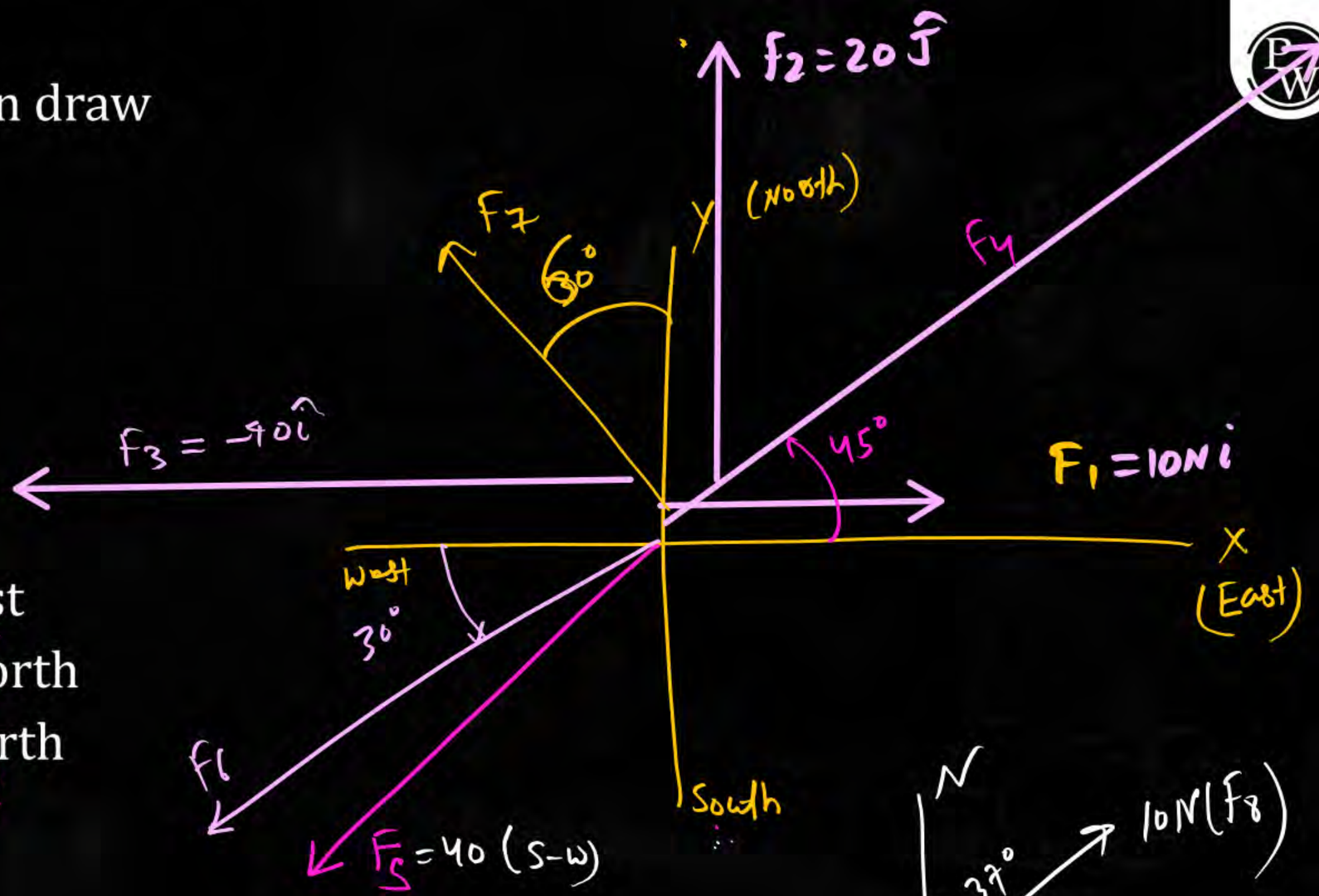
Magnitude of vector → *

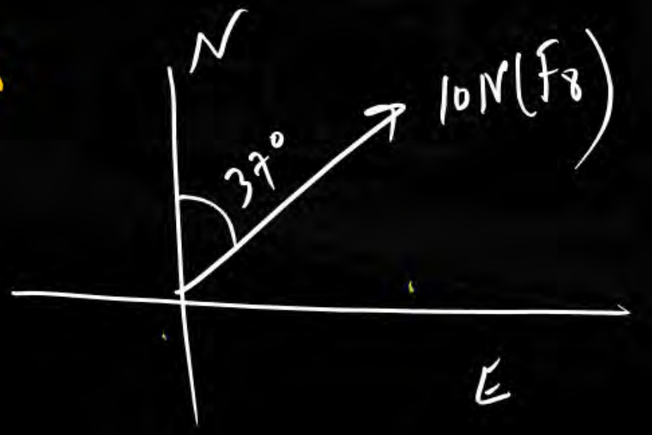
Component of vector → *

➤ Some forces are given then draw

1. $\vec{F}_1$ = 10 N along east ($\hat{i}$)
2. $\vec{F}_2$ = 20 N along north ($\hat{j}$)
3. $\vec{F}_3$ = 40 N west
4. $\vec{F}_4$ = 50 N north-east
5. $\vec{F}_5$ = 40 N south-west
6. $\vec{F}_6$ = 10 N at 30° south-west
7. $\vec{F}_7$ = 10 N at 60° west of north
8. $\vec{F}_8$ = 10 N at 37° east of north

ast 30° south





Try in rough and match Answer, no copy

➢ When vector multiplied with positive scalar then direction remains same and magnitude will change. If –ve scalar then direction reversed and magnitude change.

Q. $\vec{A} = 30\text{m}\,\hat{i}$

(i) $3\vec{A} = 3(30m\hat{i}) = 90m\hat{i}$

(ii) $\vec{A}/3 = \frac{30m\hat{i}}{3} = 10m\hat{i}$

(iii) $-5\vec{A} = -5(30m\hat{i}) = -150m\hat{i}$

(iv) $|\vec{A}| = 30m$

(v) $\hat{A} = \hat{i}$

Try in rough no copy

Question

Copy in notes (Try in rough)

If angle between vector $\vec{A}$ and $\vec{B}$ is 30° then angle between

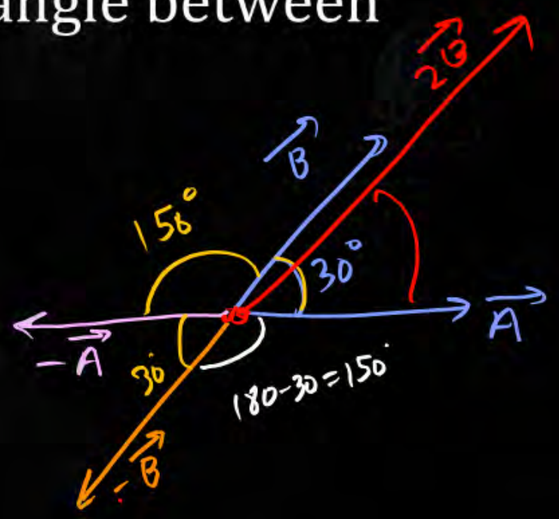
(a) $\vec{A}$ and $-\vec{B}$ 150°

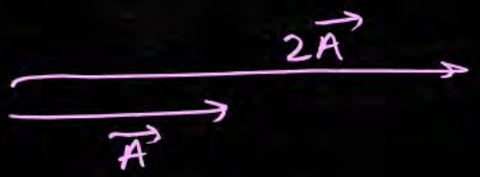
(b) $\vec{A}$ and $\vec{B}$ 30°

(c) $-\vec{A}$ and $-\vec{B}$ 30°

(d) $\vec{A}$ and $2\vec{B}$ 30°

(e) $2\vec{A}$ and $-3\vec{B}$ 150°

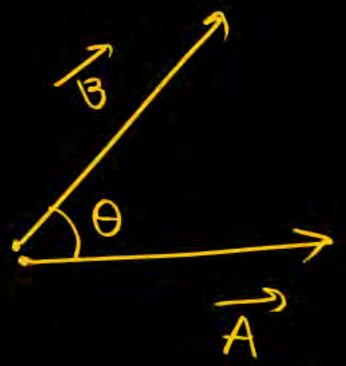


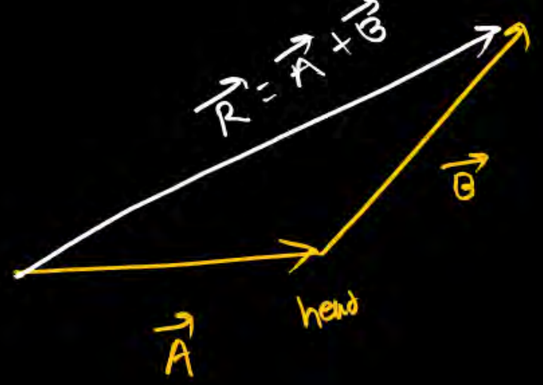


Poly gon - Law of vector addition

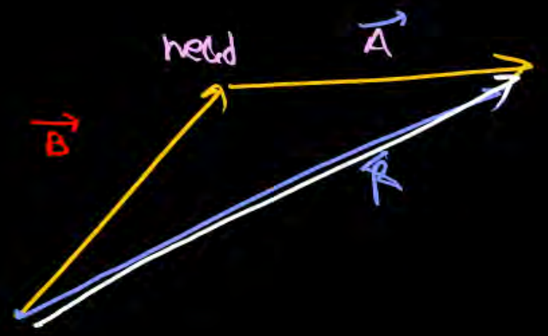
→ Koi bhi 1st vector lo, uske head se 2nd vector start karo, then 1st ke tail se 2nd ke head ko mila do; Wo vector dono ka resultant hoga.

→ graphical method of vector addition





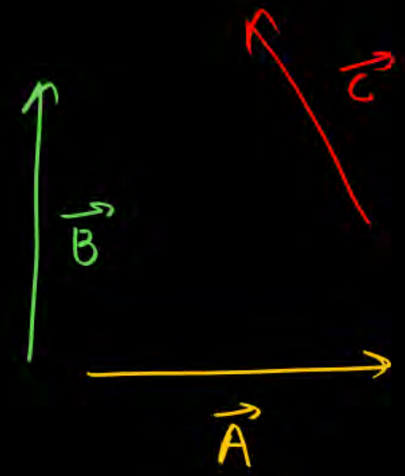
$$\vec{R} = \vec{A} + \vec{B} = \vec{B} + \vec{A}$$

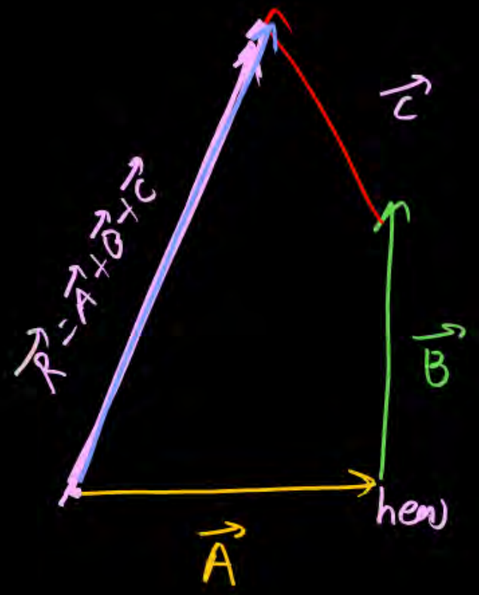


MR* BOX

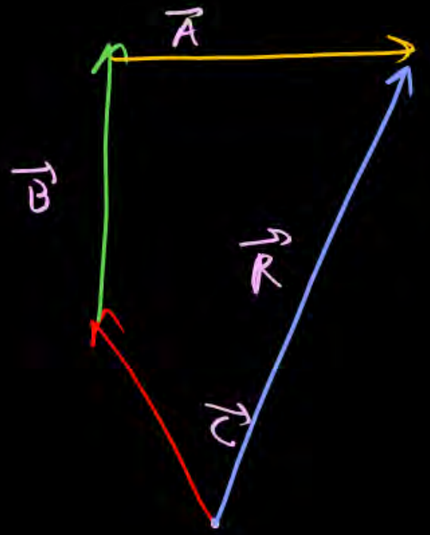
Angle nikalne ke liye tail-tail or head-head ko ek sath rakho, vector add karne ke liye 1st ke head se 2nd ka tail start karo.

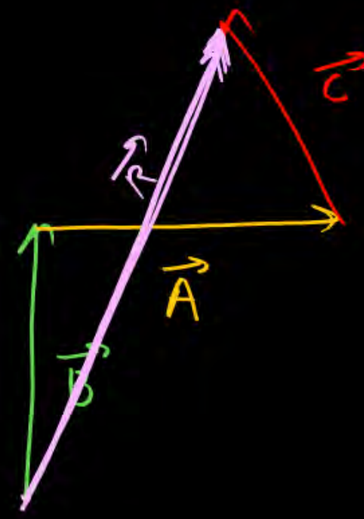
Ek bat or smajh lo jin 2-3 vector ko add kar rahe ho usme draw karne ka order kuch bhi le skte ho niche wale diagram se samjho. In all 3-diagram $\vec{R}$ is same.

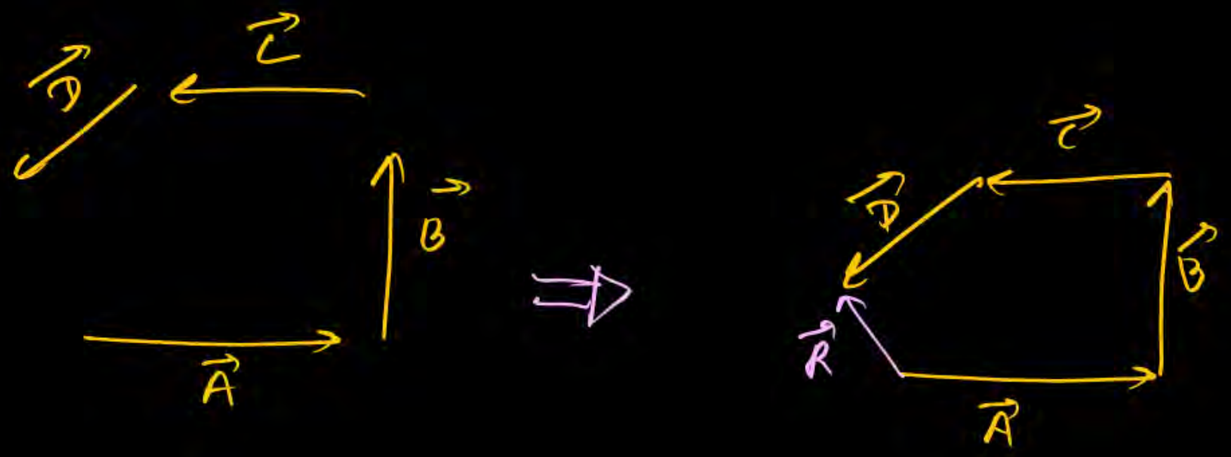




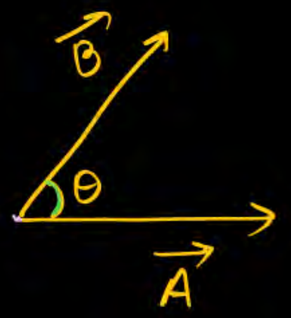
$$\vec{A}+\vec{B}+\vec{C}=\vec{B}+\vec{A}+\vec{C}=\vec{C}+\vec{B}+\vec{A}$$



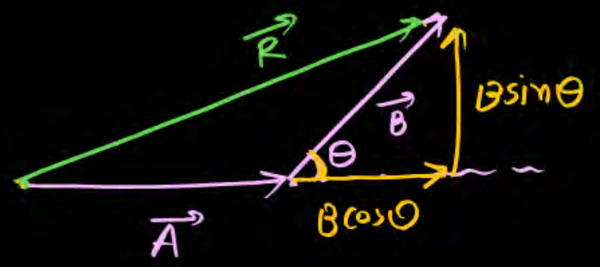



$\vec{D}$
$\vec{C}$
$\vec{B}$
$\vec{A}$
$\vec{C}$
$\vec{D}$
$\vec{B}$
$\vec{R}$
$\vec{A}$

Triangle Law of vector addition



$\Rightarrow$

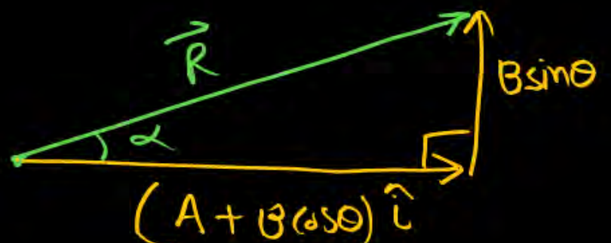


$A = |\vec{A}|$ = Magnitude of $\vec{A}$

$B = |\vec{B}|$ = magnitude of $\vec{B}$

R = Magnitude of Resultant of $\vec{A}+\vec{B}$

θ : Angle b/w $\vec{A}$ & $\vec{B}$



$$R = \sqrt{(A+B\cos\theta)^2 + (B\sin\theta)^2}$$

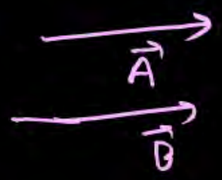
$$= \sqrt{A^2 + B^2\cos^2\theta + 2AB\cos\theta + B^2\sin^2\theta}$$

$$R = \sqrt{A^2 + B^2(\cos^2\theta + \sin^2\theta) + 2AB\cos\theta}$$

magnitude of Resultant

$$R = \sqrt{A^2 + B^2 + 2AB\cos\theta}$$

If $\theta = 0^\circ$



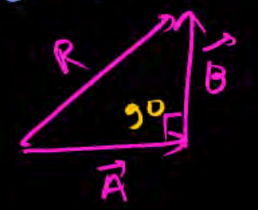
$$R = \sqrt{A^2 + B^2 + 2AB\cos 0^\circ}$$

$$R = \sqrt{(A+B)^2}$$

$$R = A + B$$

Max

$\theta = 90^\circ$



$$R = \sqrt{A^2 + B^2}$$

$\theta = 180^\circ$

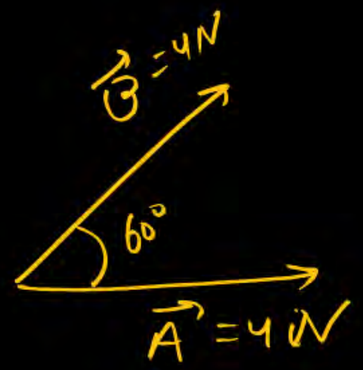
$$R = \sqrt{A^2 + B^2 - 2AB}$$

$$= \sqrt{(A-B)^2}$$

$$R = A - B$$

Min

(Q)



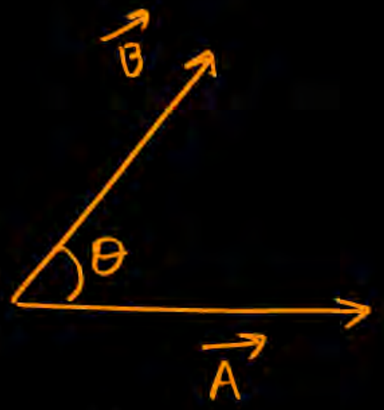
$R = ??$

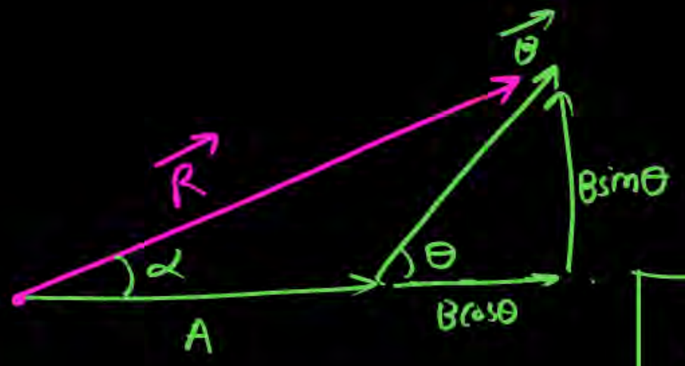
$$R = \sqrt{4^2 + 4^2 + 2 \times 4 \times 4 \times \cos 60^\circ}$$

$$R = \sqrt{16 + 16 + 2 \times 16 \times \frac{1}{2}}$$

$$= \sqrt{48} = \sqrt{16 \times 3} = \underline{4\sqrt{3}}$$

Triangle law of vector addition :-





$|\vec{A}| = 6$ & $|\vec{B}| = 4$

$R_{max} = 6+4 = 10$

$R_{min} = 6-4$
$\theta = 180^\circ \quad = 2$

Magnitude of Result

$$R = \sqrt{A^2+B^2+2AB\cos\theta}$$

- $\theta = 0^\circ$ → $R_{max} = A+B$
- $\theta = 90^\circ$ → $R = \sqrt{A^2+B^2}$
- $\theta = 180^\circ$ → $R = A-B$ (min)

$$\tan\alpha = \frac{B\sin\theta}{A+B\cos\theta}$$

↳ Angle b/w $\vec{A}$ & $\vec{R}$

$$A-B \leq R \leq A+B$$

min ↓ Results max

Question

Two vector of magnitude 3 and 4 acting at different angle. Then find their resultant:

(1) $\theta = 0°$ ⟶ $R_{max} = 3+4 = 5$

(2) $\theta = 60°$ ⟶ $R = \sqrt{3^2+4^2+2\times3\times4\cos60} = \sqrt{9+16+2\times12\times\frac{1}{2}}$

$= \sqrt{37}$

(3) $\theta = 90°$ ⟶ $R = \sqrt{(3)^2+(4)^2+2\times3\times4\cos90}$

$R = \sqrt{9+16} = 5$

(4) $\theta = 120°$ ⟶ $R = \sqrt{3^2+4^2+2\times3\times4\cos120}$

(5) $\theta = 180°$

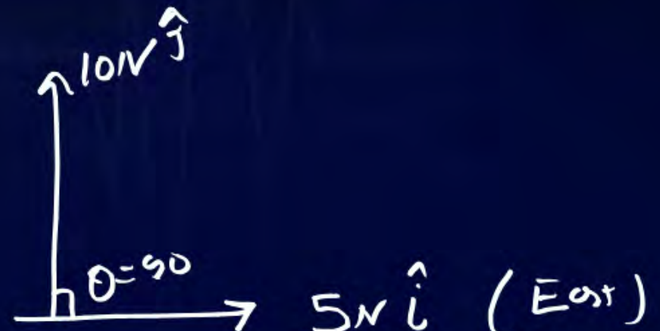
$R_m = \sqrt{3^2+4^2+2\times3\times4\cos(180)} = \sqrt{9+16-24}$

$= \sqrt{25-24} = \sqrt{1} = 1$

QUESTION

Two force $\vec{F}_1$ = 5N due east and F_2 = 10 N due north then resultant of these two force is

1. $5\sqrt{5}$ N
2. 15 N
3. 5 N
4. $\sqrt{5}$ N



$R = \sqrt{(5)^2 + (10)^2}$

$= \sqrt{125}$

$= \sqrt{25 \times 5} = 5\sqrt{5}$

QUESTION

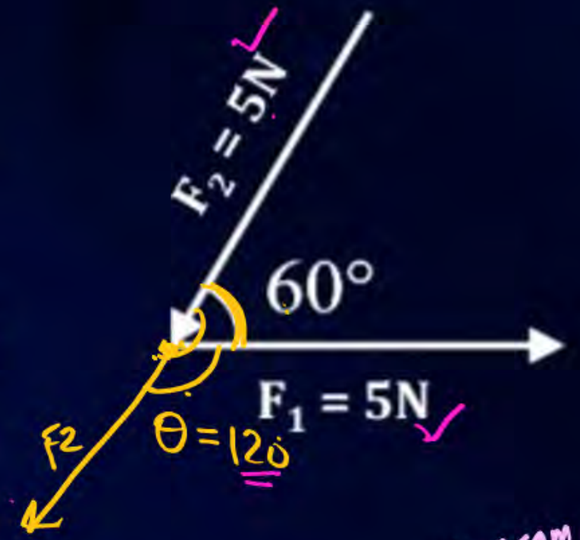
Find net force $\vec{F}_1 + \vec{F}_2$.

1. 5N
2. $5\sqrt{3}$N
3. $5\sqrt{2}$N
4. 0

$\vec{A}$ & $\vec{B}$ vect

$\vec{A} + \vec{B} = \vec{R}$ (diff ke sath)

A + B (sum of their mgnt)



$R = \sqrt{5^2 + 5^2 + 2 \times 5 \times 5 \cos 120}$

$= \sqrt{25 + 25 + 50 \times (-\frac{1}{2})}$

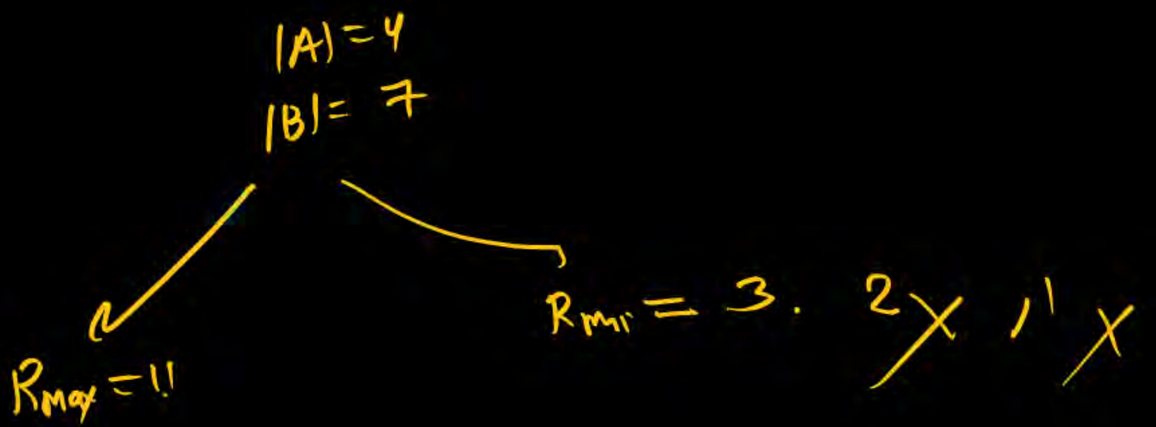
$= \sqrt{50 - 25} = \sqrt{25} = 5$

$R = \sqrt{5^2 + 5^2 + 2 \times 5 \times 5 \cos 60}$

$= \sqrt{50 + 2 \times 25 \times \frac{1}{2}} = \sqrt{75}$

NR scam

Ans: (1)


|A| = 4
|B| = 7
$R_{max} = 11$
$R_{min} = 3 \cdot 2 \times 1 \times$

Question

Which pair of two vector can not give resultant of magnitude 4.

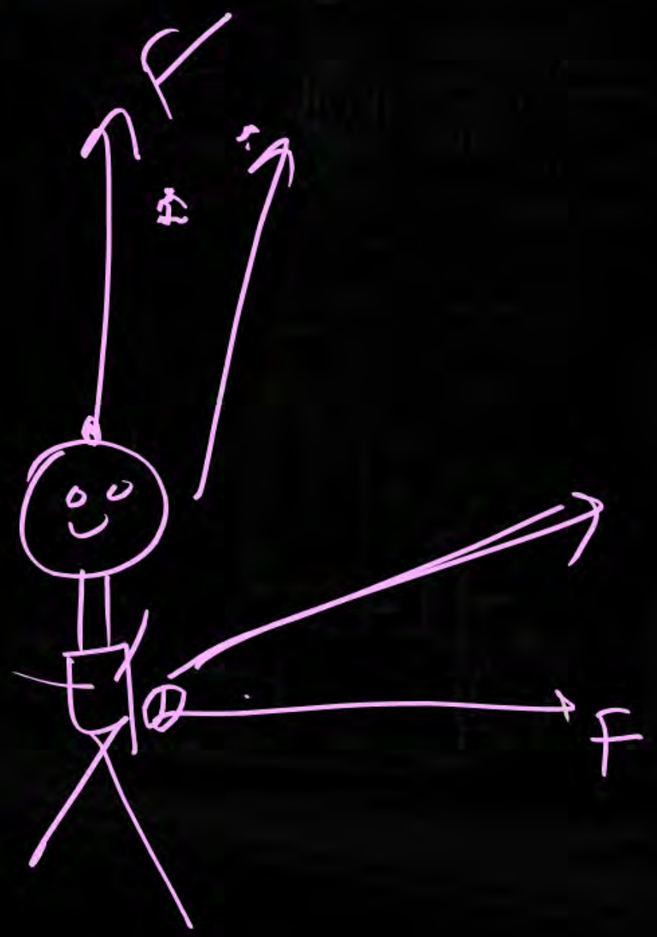
(a) 2, 4 → 4-2=2, 6 (min, max)

(b) 4, 4 → 4-4=0 , 4+4=8

(c) 4, 9 → 9-4=5, 6, 7, 8, 9, 10, 11, 12, 13 ; 14 X 15 X

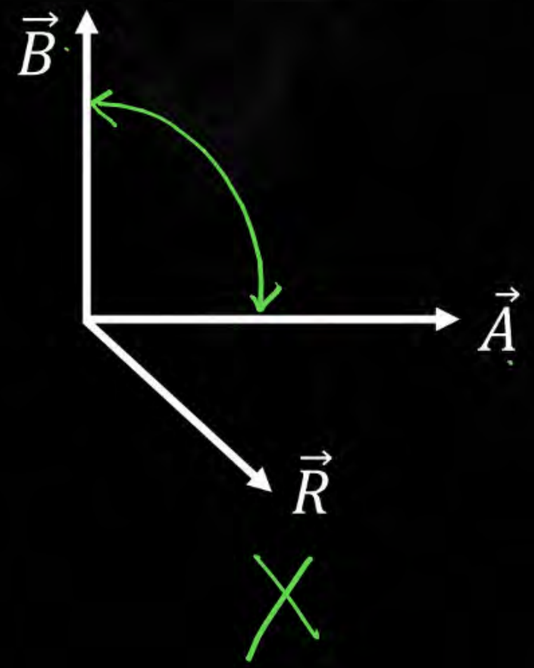
(d) 1, 5 → 5-1=4 → 5+1=6

Ans (d)

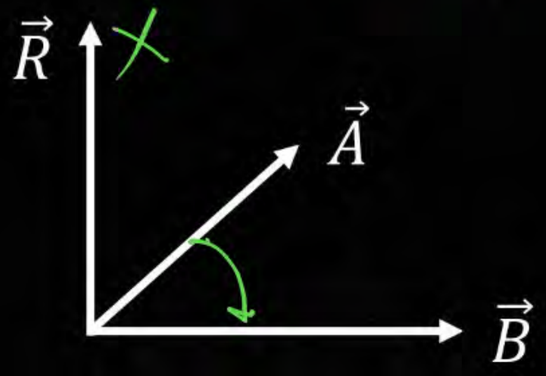


Question

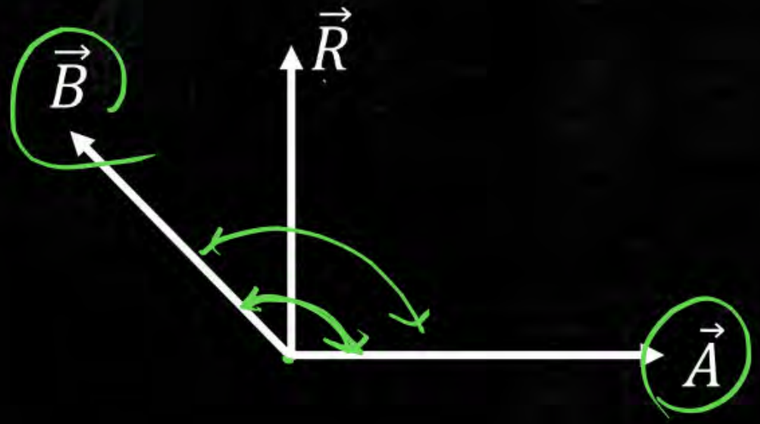
Which of the following diagram is correct: for Rein $\vec{A}+\vec{B}=\vec{R}$



X

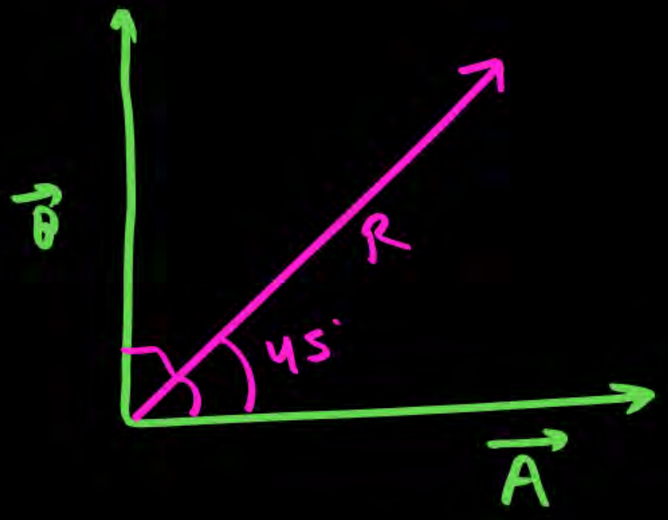


X

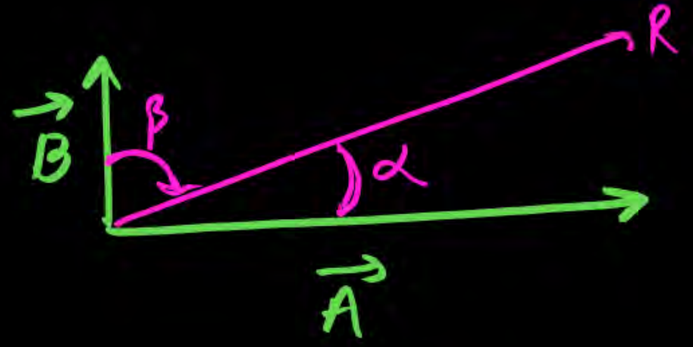


MR* BOX

2-vector ka resultant unke bich me rahega, bade wale vector ke pass niche wale diagram se feel karo



$|\vec{A}| = |\vec{B}|$



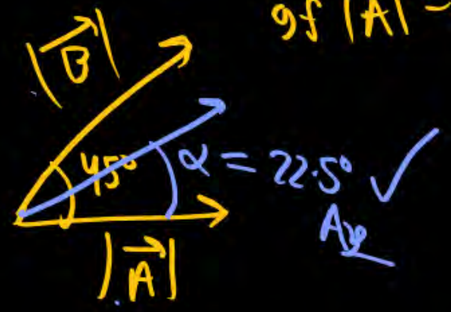
$\alpha < \beta$



$\beta < \alpha$

AIEEE

If $|\vec{A}| = |\vec{B}|$ then find Angle b/w $\vec{R}$ & $\vec{A}$.



Ans ✓

Question

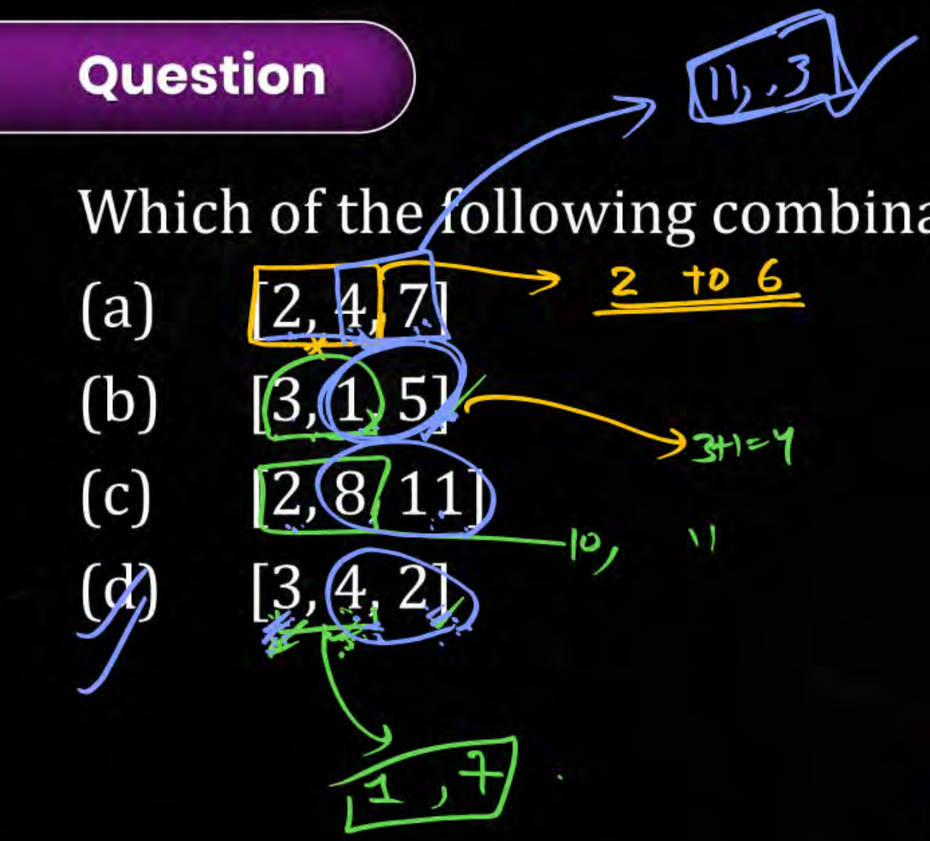
Which of the following combination of three force can give zero resultant?

(a) [2, 4, 7]

(b) [3, 1, 5]

(c) [2, 8, 11]

(d) [3, 4, 2]



Ans (d)

MR* BOX

Koi bhi 2-vector lo uske resultant ke range (A – B to A + B) me 3rd vector aaya to sum of all three vector zero ho sakta hai

If Two vector of same magnitude $|\vec{A}| = |B|$ acting at angle θ then find Their Resultant

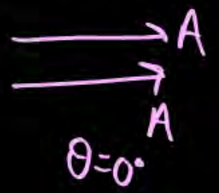
$$R = \sqrt{A^2 + A^2 + 2A \cdot A \cos\theta}$$

$$R = 2A \cos(\theta/2)$$

MR Ratta

$\theta = 0°$

$R = 2A$

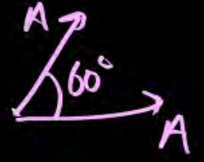


$\theta = 60°$

$R = 2A \cos(30)$

$= 2A \frac{\sqrt{3}}{2}$

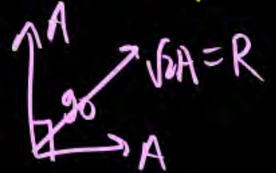
$= \sqrt{3}A$



$\theta = 90°$

$R = 2A \cos(45)$

$= 2A \times \frac{1}{\sqrt{2}} = \sqrt{2}A$



$\theta = 120°$

$R = 2A \frac{1}{2}$

$R = A$



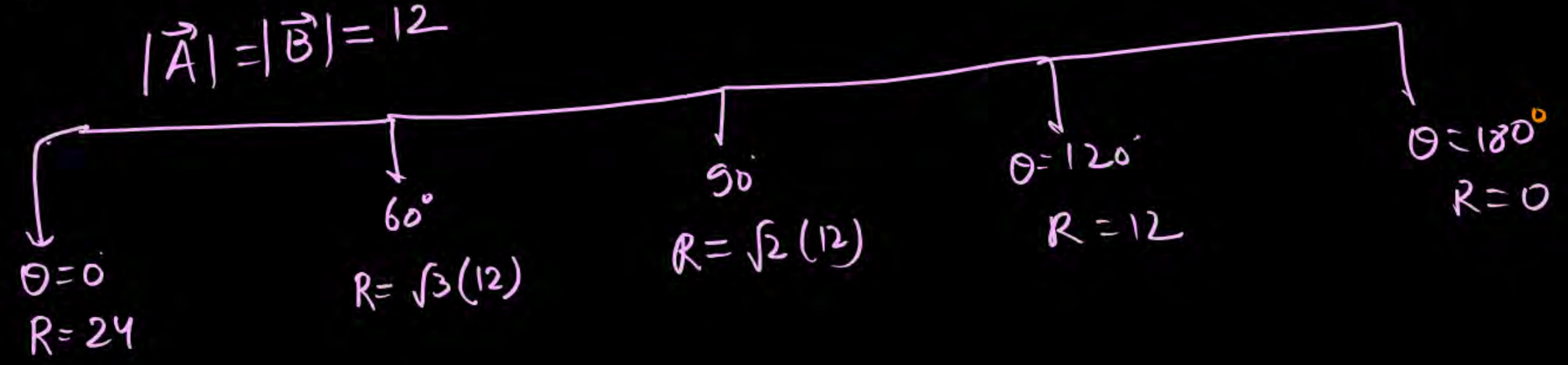
$\theta = 180°$

$R = 0$

$R = 0$

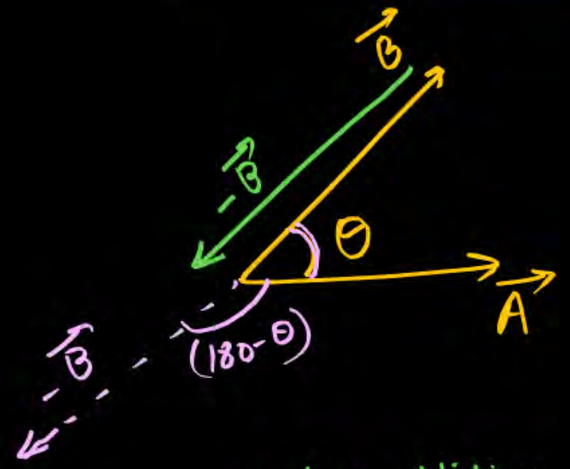


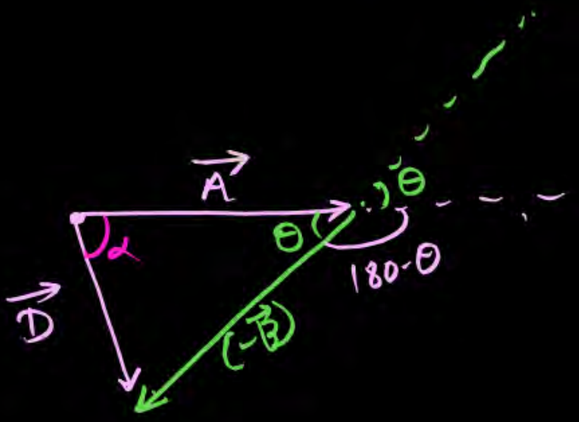
#


$|\vec{A}| = |\vec{B}| = 12$
$\theta = 0$
$R = 24$
$60°$
$R = \sqrt{3}(12)$
$90°$
$R = \sqrt{2}(12)$
$\theta = 120°$
$R = 12$
$\theta = 180°$
$R = 0$

Vector subtraction

$$\vec{A} - \vec{B} = \vec{A} + (-\vec{B})$$





$$D = \sqrt{A^2 + B^2 + 2AB\cos(180-\theta)}$$

$$D = \sqrt{A^2 + B^2 - 2AB\cos\theta}$$

$$\boxed{D = \sqrt{A^2 + B^2 - 2AB\cos\theta}}$$

$\theta = 0$

$$D = \sqrt{A^2 + B^2 - 2AB \times 1}$$

$$= \sqrt{(A-B)^2}$$

$$\boxed{D = A - B \quad \text{min}}$$

$\theta = 90°$

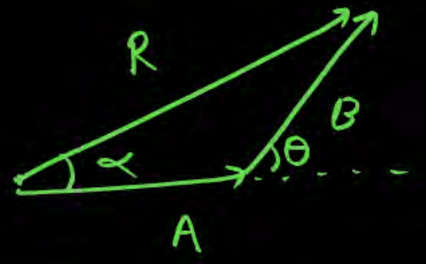
$$D = \sqrt{A^2 + B^2}$$

$\theta = 180$

$$D = A + B$$

180°

Vector addition



$$\tan\alpha = \frac{B\sin\theta}{A + B\cos\theta}$$

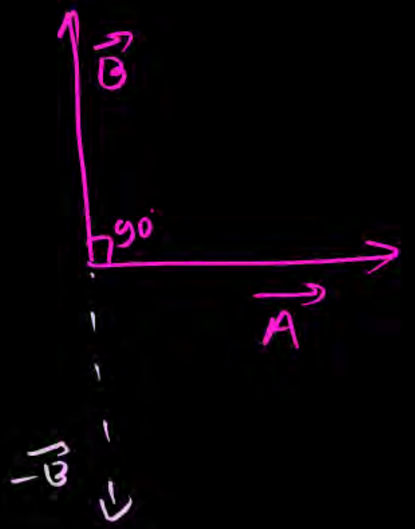
$$R = \sqrt{A^2 + B^2 + 2AB\cos\theta}$$

$$R = 2A\cos(\theta/2)$$

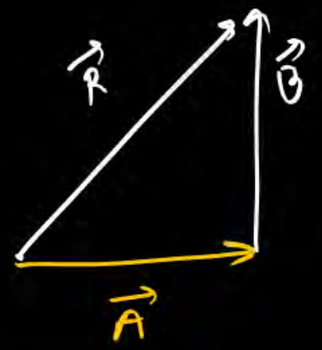
$$\cos(180-\theta) = -\cos\theta$$



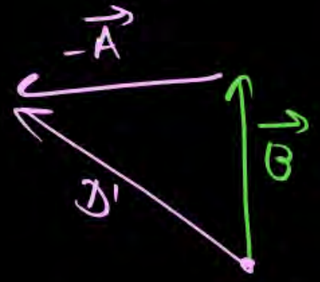
$$\tan\alpha = \frac{B\sin\theta}{A - B\cos\theta}$$



$\vec{R} = \vec{A} + \vec{B}$

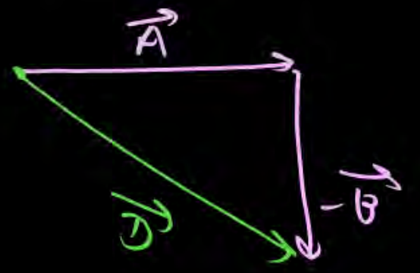


$\vec{D'} = \vec{B} - \vec{A} = \vec{B} + (-\vec{A})$



- $\vec{A} + \vec{B} = \vec{B} + \vec{A}$
- $\vec{A} - \vec{B} \neq \vec{B} - \vec{A}$

$\vec{D} = \vec{A} - \vec{B} = \vec{A} + (-\vec{B})$



Two vector of same magnitude $|\vec{A}| = |\vec{B}| = A$ acting at angle θ then. magnitude of their Resultant & magnitude of their Subtraction.

$R = 2A\cos\theta/2$ $\quad$ $D = 2A\sin(\theta/2)$

$\theta = 0^\circ$	$\theta = 60^\circ$	$\theta = 90^\circ$	$\theta = 120^\circ$	$\theta = 180^\circ$
$R = 2A$	$R = \sqrt{3}A$	$R = \sqrt{2}A$	$R = A$ ✓	$R = 0$
$D = 0$	$D = A$	$D = \sqrt{2}A$	$D = \sqrt{3}A$ ✓	$D = 2A$

m_R

Vector Addition

2016, 2006, 2001

If $\vec{A} + \vec{B} = \vec{R}$ and $\vec{D} = \vec{A} - \vec{B}$ then angle between $\vec{A}$ + and $\vec{B}$ for given condition

$|\vec{A}| = |\vec{B}| = |\vec{R}| \rightarrow 120°$

$|\vec{A}| = |\vec{B}| = |\vec{D}| \rightarrow 60°$

$|\vec{A} + \vec{B}| = |\vec{A} - \vec{B}| \rightarrow 90°$

$R = \sqrt{A^2 + B^2 + 2AB\cos\theta} \rightarrow R = 2A\cos(\theta/2)$

$D = \sqrt{A^2 + B^2 - 2AB\cos\theta} \rightarrow D = 2A\sin(\theta/2)$

If the magnitude of sum of two vectors is equal to the magnitude of difference of the two vectors, the angle between these vectors is: **(2016-I)**

1. 1°
2. 90°
3. 45°
4. 180°

Question

If $|\vec{A}+\vec{B}| = |\vec{A}| + |\vec{B}|$ then angle between A and B will be: (2001)

$R = A + B$

1. $90°$
2. $120°$
3. $0°$
4. $60°$

Question

For Two vector $\vec{A}$ and $\vec{B}$, vector addition $\vec{R} = \vec{A} + \vec{B}$ and vector Subtraction $\vec{D} = \vec{A} - \vec{B}$ then Angle between $\vec{A}$ and $\vec{B}$ for given condition.

(1) $|\vec{A} + \vec{B}| = |\vec{R}| = |\vec{A}| = |\vec{B}|$ → 120°

(2) $|\vec{A}| = |\vec{B}| = |\vec{A} - \vec{B}|$ → 60°

(3) $A^2 + B^2 = |\vec{A} - \vec{B}|^2 = D^2$ → $\sqrt{A^2+B^2} = D$ (90°)

(4) $A^2 + B^2 = R^2$ → $\theta = 90°$

(5) $|\vec{A} + \vec{B}| = |\vec{A} - \vec{B}|$ → 90° ✓

(B) $|\vec{A} + \vec{B}| = R = |\vec{A}| + |\vec{B}|$ $R = A + B \Rightarrow 0°$

(7) $D = |\vec{A} - \vec{B}| = |\vec{A}| + |\vec{B}|$ → $D = A + B$ → 180°

(8) $|\vec{A} + \vec{B}| = R = |\vec{A}| - |\vec{B}|$ → $R = A - B$ → 180°

(9) $|\vec{A} - \vec{B}| = D = |\vec{A}| - |\vec{B}|$ → $\theta = 0°$

(10) $R = |\vec{A} + \vec{B}| = \sqrt{3}A = \sqrt{3}B$ → $\theta = 60°$

(11) $|\vec{A} - \vec{B}| = \sqrt{3}A = \sqrt{3}B$ → (120°)

Question

If the sum of two unit vectors is a unit vector, then the magnitude of their difference is:

$|\vec{A}| = 1$ ✓
$|\vec{B}| = 1$ ✓
$\longrightarrow$ $|\vec{A}+\vec{B}| = |\vec{R}| = 1$ ✓
$\theta = 120°$ ✓

$|D| = |\vec{A} - \vec{B}| = \sqrt{3} \times 1$

$D = \sqrt{1^2 + 1^2 - 2 \times 1 \times 1 \cos 120°}$

$\sqrt{3}$

Question

finding of unit vector :—

Find unit vector $\hat{A} = \frac{\vec{A}}{A}$

(1) $\vec{A} = \hat{\imath}$ ⟶ $\hat{A} = \hat{\imath}$

(2) $\vec{B} = \hat{\imath} + \hat{\jmath}$ $\hat{B} = \frac{\hat{\imath}+\hat{\jmath}}{\sqrt{2}}$

(3) $\vec{C} = \hat{\imath} + \hat{\jmath} - \hat{k}$ $\hat{C} = \frac{\hat{\imath}+\hat{\jmath}-\hat{k}}{\sqrt{3}}$

(4) $\vec{D} = \sin\theta\,\hat{\imath} - \cos\hat{\jmath}$ $\vec{D} = \sin\theta\,\hat{\imath} - \cos\theta\,\hat{\jmath}$

(5) $\vec{E} = 3\hat{\imath} + 4\hat{\jmath}$ $\vec{E} = \frac{3\hat{\imath}+4\hat{\jmath}}{5}$

$\vec{A} = A\hat{A}$

$\hat{A} = \frac{\vec{A}}{A}$ ✓

Unit vector $\hat{A} = \frac{\vec{A}}{|A|}$.

- Vector addition / subtraction in terms of component:

$\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}$ and $\vec{B} = B_x\hat{i} + B_y\hat{j} + B_z\hat{k}$

$\vec{A} + \vec{B} = (A_x + B_x)\hat{i} + (A_y + B_y)\hat{j} + (A_z + B_z)\hat{k}$

$\vec{A} - \vec{B} = (A_x\hat{i} + A_y\hat{j} + A_z\hat{k}) - (B_x\hat{i} + B_y\hat{j} + B_z\hat{k}) = (A_x - B_x)\hat{i} + (A_y - B_y)\hat{j} + (A_z - B_z)\hat{k}$

$\vec{B} - \vec{A} = (B_x - A_x)\hat{i} + (B_y - A_y)\hat{j} + (B_z - A_z)\hat{k}$

$|\vec{A}| + |\vec{B}|$

↑ Sum of magnitude

Question

$\vec{A} = 4\hat{i} + 3\hat{j}; \vec{B} = 3\hat{i} - 4\hat{j}$ then find

(1) $\vec{A} + \vec{B} =$ $(4i+3\hat{j}) + (3i-4\hat{j}) = 7i - \hat{j}$

(2) $\vec{A} - \vec{B} =$ $\hat{i} + 7\hat{j}$

(3) $\vec{B} - \vec{A} =$ $3i-4\hat{j} - 4i-3\hat{j} = -i-7\hat{j}$

(4) $|\vec{A}| + |\vec{B}| =$ $5+5 = 10$

Question

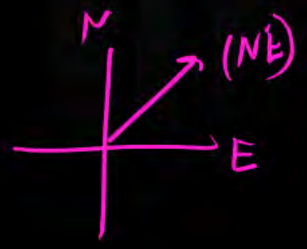
Initial velocity V = $10\hat{i}$ (in east) after some time its velocity becomes $10\hat{j}$ m/s north then find change in velocity is?

$\vec{V_i} = 10\hat{i}$

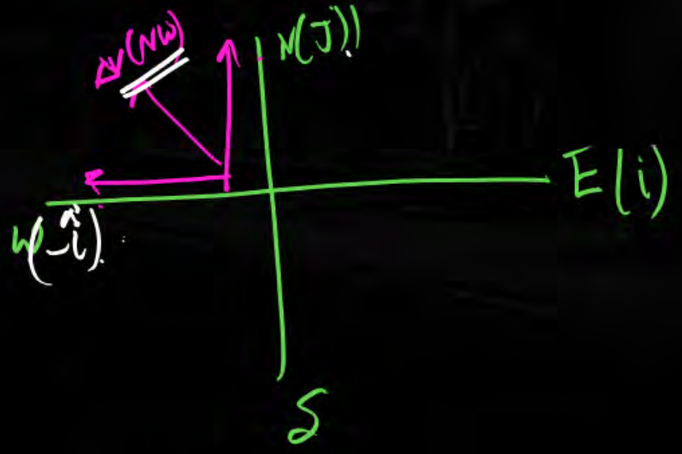
$\vec{V_f} = 10\hat{j}$

Sum of velocity $= \vec{V_i} + \vec{V_f}$

$= 10\hat{i} + 10\hat{j}$



Change in velocity $\vec{V_f} - \vec{V_i} = 10\hat{j} - 10\hat{i}$



Question

Two vector $\vec{A} = 6\hat{i}$ and $\vec{B} = 8\hat{j}$ then

(1) $\vec{A} + \vec{B} =$ $6i + 8j$

(2) $\vec{A} - \vec{B} =$ $6i - 8j$

(3) $|\vec{A}| + |\vec{B}| =$ $6 + 8 = 14$

(4) $|\vec{A} + \vec{B}| =$??

$$|\vec{A}+\vec{B}| = |6i+8\vec{j}| = \sqrt{6^2+8^2}$$

$$= \sqrt{36+64} = 10$$

Question

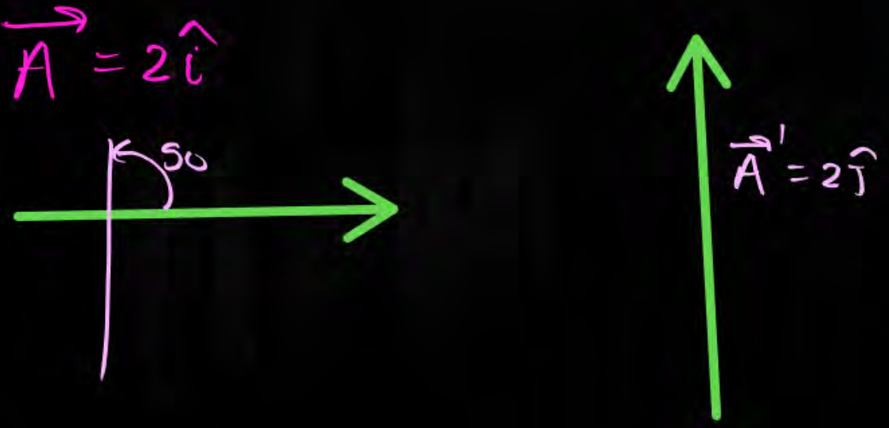
A vector $2\hat{i}$ rotated, then new vector will be:

(i) By $\theta = 90°$ (A.C) = $A' = 2\hat{j}$

(ii) By $\theta = 180°$ (A.C) = $A' = -2\hat{i}$

(iii) By $\theta = 270°$ (A.C) = $A' = -2\hat{j}$

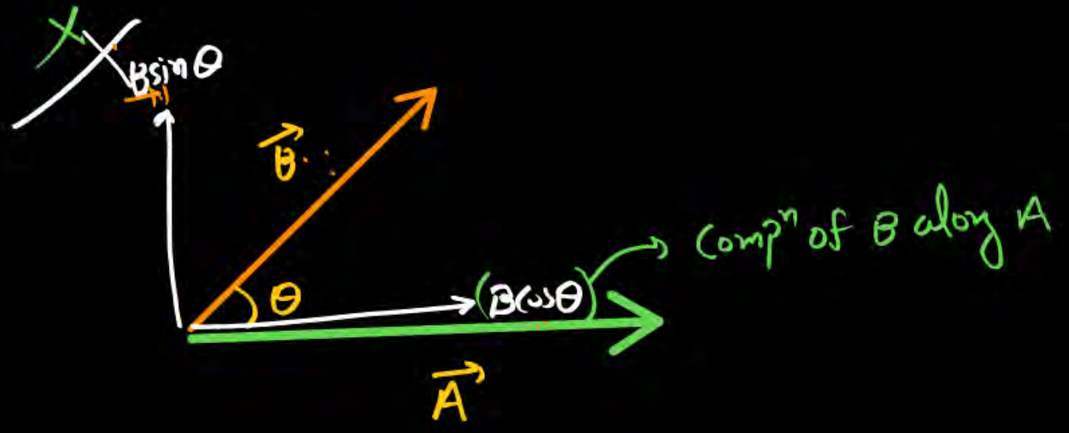
(iv) By $\theta = 360°$ (A.C) = $A' = 2\hat{i}$



MR* BOX

In rotation of vector magnitude remains same and direction changes.

Dot Product [Scalar Product]



$$\vec{A} \cdot \vec{B} = A B \cos\theta$$

θ → Angle b/w $\vec{A}$ & $\vec{B}$

$\vec{A}$ → vector

$$\begin{bmatrix} \hat{i}\cdot\hat{i} = 1\times 1\cos 0 = 1 \\ j\cdot j = 1 \\ k\cdot k = 1 \end{bmatrix} \qquad \begin{bmatrix} i\cdot j = 0 \\ i\cdot k = 0 \\ k\cdot j = 0 \end{bmatrix}$$

(iv) To check unit vector

$$\vec{A}\cdot\vec{A} = 1$$

$$\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}$$

$$\vec{B} = B_x\hat{i} + B_y\hat{j} + B_z\hat{k}$$

$$\left\{ \vec{A}\cdot\vec{B} = A_xB_x + A_yB_y + A_zB_z \right\}$$

Application of dot Product :-

To check two vector

(i) If $\vec{A}\cdot\vec{B} = 0$ hence $\vec{A} \perp \vec{B}$

(ii) To find Angle b/w vector

$$\frac{\vec{A}\cdot\vec{B}}{AB} = \cos\theta$$

(iii) To find component of one vector along other.

$$\vec{A}\cdot\vec{B} = A(B\cos\theta)$$

$$B\cos\theta = \frac{\vec{A}\cdot\vec{B}}{A} = \text{Comp}^{n} \text{ of } B \text{ along } \vec{A}$$

Question

If $\vec{A} = 2\hat{i} + 3\hat{j} + \alpha\hat{k}$ and $\vec{B} = \hat{i} - 2\hat{j} + 4\hat{k}$ find α. If $\vec{A}$ is perpendicular to $\vec{B}$. (NEET)

$$2 - 6 + 4\alpha = 0$$

$$-4 + 4\alpha = 0$$

$$4\alpha = 4$$

$$\alpha = 1$$

QUESTION

If a vector $2\hat{i} + 3\hat{j} + 8\hat{k}$ is perpendicular to the vector $3\hat{i} - 4\hat{j} + \alpha\hat{k}$, then the value of α is [2005]

1. 1/2
2. -1/2
3. 3/4
4. -1

$\vec{A} \cdot \vec{B}$

$6 - 12 + 8\alpha = 0$

$-6 + 8\alpha = 0$

$8\alpha = 6$

$\alpha = \frac{6}{8} = \frac{3}{4}$

QUESTION

If vector $\vec{A} = \cos \omega t\, \hat{i} + \sin \omega t\, \hat{j}$ and $\vec{B} = \cos \frac{\omega t}{2}\, \hat{i} + \sin \frac{\omega t}{2}\, \hat{j}$ are function of time, then the value of t is at which they are orthogonal to each other is [2005]

Perpendicular

1. $t = \pi/\omega$
2. $t = 0$
3. $t = \pi/4\omega$
4. $t = \pi/2\omega$

$$\cos(\omega t) \cdot \cos\left(\frac{\omega t}{2}\right) + \sin(\omega t) \cdot \sin\left(\frac{\omega t}{2}\right) = 0$$

$$\cos(A - B) = \cos\left(\omega t - \frac{\omega t}{2}\right) = 0$$

$$\cos\left(\frac{\omega t}{2}\right) = 0$$

$$\frac{\omega t}{2} = \frac{\pi}{2}$$

$$t = \pi/\omega$$

Ans: (1)

QUESTION

The angle between the two vectors $\vec{A} = 3\hat{i} + 4\hat{j} + 5\hat{k}$ and $\vec{B} = 3\hat{i} + 4\hat{j} - 5\hat{k}$ will be [1994]

1. $90°$
2. $180°$
3. zero
4. $45°$

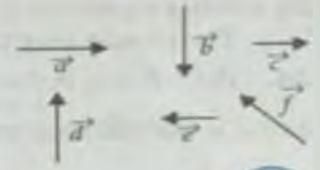
Question

If $\vec{A} = 2\hat{\imath} + 6\hat{\jmath} + 3\hat{k}$ and $\vec{B} = 4\hat{\imath}$. Find angle between $\vec{A}$ and $\vec{B}$.

$$\vec{A} \cdot \vec{B} = A \cdot B \cos\theta$$

3.4 Addition and Subtraction of Vectors-Graphical Method

2. Six vectors, $\vec{a}$ through $\vec{f}$ have the magnitudes and directions indicated in the figure. Which of the following statements is true?



(a) $\vec{b}+\vec{c}=\vec{f}$ (b) $\vec{d}+\vec{c}=\vec{f}$
(c) $\vec{d}+\vec{e}=\vec{f}$ (d) $\vec{b}+\vec{e}=\vec{f}$ *(2010)*

3.5 Resolution of Vectors

3. If a unit vector is represented by $0.5\hat{i}-0.8\hat{j}+c\hat{k}$ then the value of c is
(a) $\sqrt{0.01}$ (b) $\sqrt{0.11}$
(c) 1 (d) $\sqrt{0.39}$ *(1999)*

3.6 Vector Addition-Analytical Method

4. If the magnitude of sum of two vectors is equal to the magnitude of difference of the two vectors, the angle between these vectors is
(a) 45° (b) 180°
(c) 0° (d) 90° *(NEET-I 2016)*

5. The vectors $\vec{A}$ and $\vec{B}$ are such that $|\vec{A}+\vec{B}| = |\vec{A}-\vec{B}|$. The angle between the two vectors is
(a) 45° (b) 90°
(c) 60° (d) 75° *(2006, 1996, 1991)*

6. If $|\vec{A}+\vec{B}| = |\vec{A}|+|\vec{B}|$ then angle between A and B will be
(a) 90° (b) 120° (c) 0° (d) 60°. *(2001)*

7. The magnitude of vectors $\vec{A}$, $\vec{B}$ and $\vec{C}$ are 3, 4 and 5 units respectively. If $\vec{A}+\vec{B}=\vec{C}$, the angle between $\vec{A}$ and $\vec{B}$ is
(a) $\pi/2$ (b) $\cos^{-1}(0.6)$
(c) $\tan^{-1}(7/5)$ (d) $\pi/4$. *(1988)*

$\sqrt{(0.5)^2+(0.8)^2+c^2}=1$

$0.25+0.64+c^2=1$

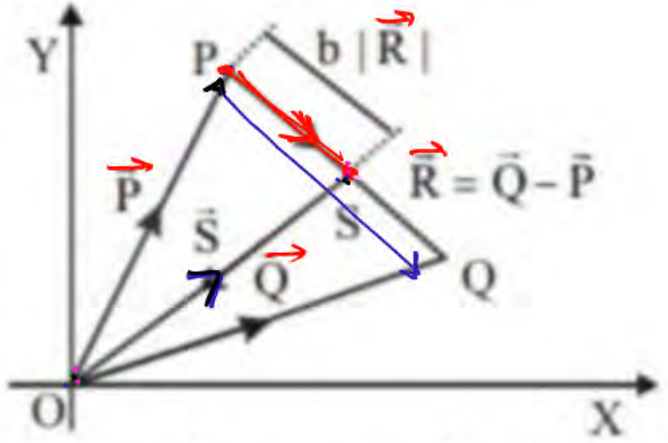
$\vec{A}=0.5\hat{i}-0.8\hat{j}+c\hat{k}$

$\vec{A}=0.5\hat{i}-0.8\hat{j}+c\hat{k}$

$\vec{A}\cdot\vec{A}=1$

$5=\sqrt{3^2+4^2+2\times3\times4\cos\theta}$

5. Three vectors $\vec{P}, \vec{Q}$ and $\vec{R}$ are shown in the figure. Let S be any point on the vector $\vec{R}$. The distance between the points P and S is $b|\vec{R}|$. The general relation among vectors $\vec{P}, \vec{Q}$ and $\vec{S}$ is **[Adv. 2017]**



(a) $\vec{S} = (1-b)\vec{P} + b\vec{Q}$ (b) $\vec{S} = (b-1)\vec{P} + b\vec{Q}$

(c) $\vec{S} = (1-b^2)\vec{P} + b\vec{Q}$ (d) $\vec{S} = (1-b)\vec{P} + b^2\vec{Q}$

$\vec{P} + b|\vec{R}|\hat{R} = \vec{S}$

$\vec{P} + b|\vec{R}|\frac{\vec{R}}{|\vec{R}|} = \vec{S}$

$\vec{P} + b\vec{R} = \vec{S}$

$\vec{P} + b(\vec{Q} - \vec{P}) = \vec{S}$

$\vec{P} + b\vec{Q} - b\vec{P} = \vec{S}$

$\vec{S} = \vec{P}(1-b) + b\vec{Q}$

Two vectors $\vec{A}$ and $\vec{B}$ are defined as $\vec{A} = a\,i$ and $\vec{B} = a\,(\cos \omega t\, \hat{i} + \sin \omega t\, \hat{j})$, where a is a constant and $\omega = \pi/6$ rad s^{-1}. If $|\vec{A} + \vec{B}| = \sqrt{3}\,|\vec{A} - \vec{B}|$ at time $t = \tau$ for the first time, the value of τ, in seconds, is _____

[JEE(Advance) 2018]

$\vec{A}\cdot\vec{B} = AB\cos\theta$

$a^2\cos(\omega t) = a \times a \times \cos 60$

$\omega t = \pi/3$

$\frac{\pi}{6} t = \pi/3$

$t = 2$ s

(#) $\vec{A} = a\hat{i}$ $\quad |\vec{A}| = a$

$\vec{B} = a\cos(\omega t)\hat{i} + a\sin(\omega t)\hat{j}$

(#) $|\vec{B}| = \sqrt{a^2(\cos^2\omega t + \sin^2(\omega t))} = \sqrt{a} = a$

$|\vec{A}+\vec{B}| = \sqrt{3}\,|\vec{A}-\vec{B}|$

$2a\cos(\theta/2) = \sqrt{3}\,2a\sin\theta/2$

$\tan(\theta/2) = \frac{1}{\sqrt{3}}$

$30^\circ = \frac{\theta}{2}$

$\theta = 60$

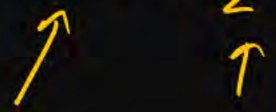
Question

JEE

If resultant of two vector is half of bigger vector and perpendicular to smaller vector then find angle between the vector.

Question

Two forces having magnitude A and $\frac{A}{2}$ are perpendicular to each other. The magnitude of their resultant is: **[JEE Mains 2023]**



1. $\frac{\sqrt{5}A}{4}$
2. $\frac{\sqrt{5}A}{2}$
3. $\frac{5A}{2}$
4. $\frac{\sqrt{5}A^2}{2}$

$$R = \sqrt{A^2 + (A/2)^2 + 2A \cdot \frac{A}{2} \cdot \cos 90^\circ}$$

$$= \sqrt{A^2 + \frac{A^2}{4} + 0}$$

$$= \sqrt{\frac{5A^2}{4}}$$

$$= \frac{A}{2}\sqrt{5}$$

Question

When vector $\vec{A} = 2\hat{i} + 3\hat{j} + 2\hat{k}$ is subtracted from vector $\vec{B}$ it gives a vector equal to $2\hat{j}$. Then the magnitude of vector $\vec{B}$ will be:

[JEE Mains 2023]

1. $\sqrt{5}$
2. 3
3. $\sqrt{6}$
4. $\sqrt{33}$

$\vec{B} - \vec{A} = 2\hat{j}$

$\vec{B} = 2\hat{j} + \vec{A}$

$\vec{B} = 2\hat{j} + 2\hat{i} + 3\hat{j} + 2\hat{k}$

$\vec{B} = 2\hat{i} + 5\hat{j} + 2\hat{k}$

$|\vec{B}| = \sqrt{4 + 25 + 4}$

$= \sqrt{33}$

Question

An electric field is given by $(6\hat{i} + 5\hat{j} + 3\hat{k})$ N C^{-1}. The electric flux through a surface area $30\hat{i}$ m^2 lying in YZ-plane (in SI unit) is:

[JEE Mains 2024]

1. 90
2. 150
3. 180
4. 60

$\vec{E} = 6\hat{i} + 5\hat{j} + 3\hat{k}$

$\vec{A} = 30\hat{i}$

$\phi = \vec{E} \cdot \vec{A}$

$= 180$

Question

If two vectors $\vec{A}$ and $\vec{B}$ having equal magnitude R are inclined at an angle θ, then

[JEE Mains 2024]

1. $\vec{A} - \vec{B} = \sqrt{2}R \sin\frac{\theta}{2}$

2. $\vec{A} + \vec{B} = 2R \sin\frac{\theta}{2}$

3. $\vec{A} + \vec{B} = 2R \cos\frac{\theta}{2}$

4. $\vec{A} - \vec{B} = 2R \cos\frac{\theta}{2}$

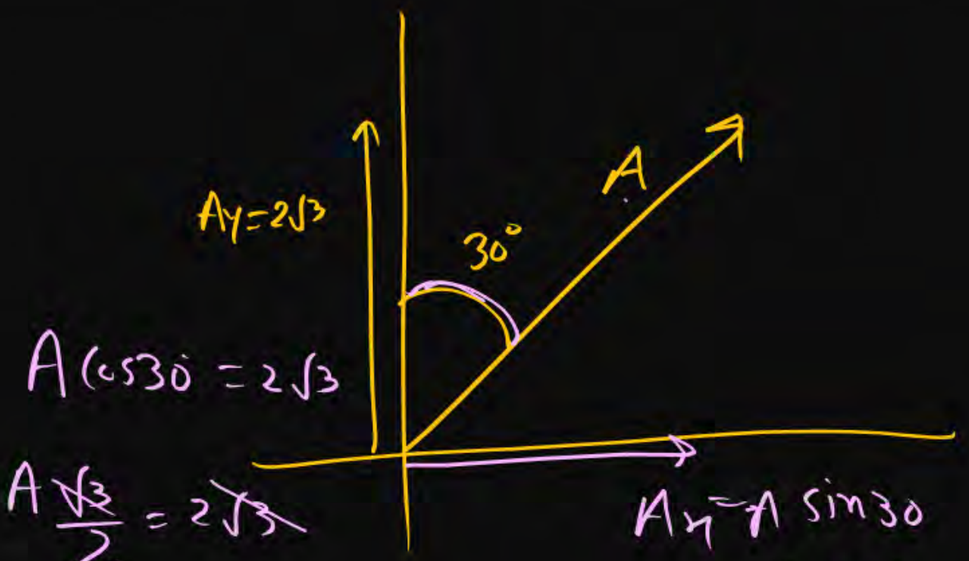
$|\vec{A}| = |\vec{B}| = R$

Question

A vector in x-y plane makes an angle of 30° with y-axis. The magnitude of y-component of vector is $2\sqrt{3}$. The magnitude of x-component of the vector will be:

[JEE Mains 2023]

1. $1/\sqrt{3}$
2. 6
3. 2
4. $\sqrt{3}$



Question

If two vectors $\vec{P} = \hat{i} + 2m\hat{j} + m\hat{k}$ and $\vec{Q} = 4\hat{i} - 2\hat{j} + m\hat{k}$ are perpendicular to each other. Then, the value of m will be

[JEE Mains 2023]

1. -1
2. 2
3. 3
4. 1

$$4 - 4m + m^2 = 0$$

$$m^2 - 4m + 4 = 0$$

$$m^2 - 2m - 2m + 4 = 0$$

Question

If the projection of $2\hat{i} + 4\hat{j} - 2\hat{k}$ on $\hat{i} + 2\hat{j} + \alpha\hat{k}$ is zero. Then, the value of α will be ____.

Component

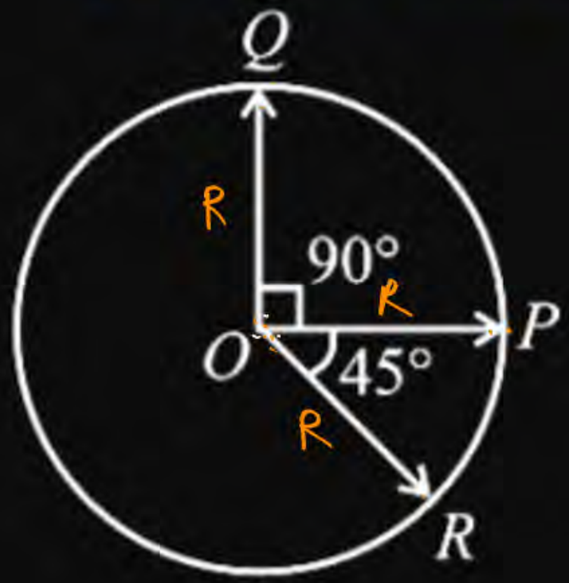
[JEE Mains 2021/22]

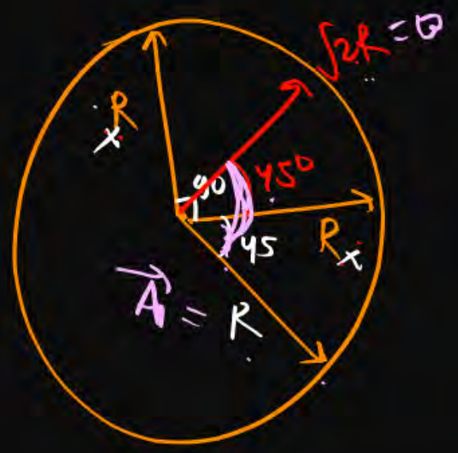
$\vec{A} \perp_r \vec{B}$

$\vec{A} \cdot \vec{B} = 0$

Question

Three vectors $\overrightarrow{OP}, \overrightarrow{OQ}$ and $\overrightarrow{OR}$ each of magnitude A are acting as shown in figure. The resultant of the three vectors is $A\sqrt{x}$. The value of x is _____. **[JEE Mains 2024]**





$x = 3$

$R_{net} = \sqrt{R^2 + 2R^2 + 2 \times R \times \sqrt{2}R \cos 90^\circ}$

$R\sqrt{x} = \sqrt{3R^2}$

Question

A small particle moves to position $5\hat{i} - 2\hat{j} + \hat{k}$ from its initial position $2\hat{i} + 3\hat{j} - 4\hat{k}$ under the action of force $5\hat{i} + 2\hat{j} + 7\hat{k}$ N. The value of work done will be _____ J.

[JEE Mains 2023]

final

initial

$W = \vec{F} \cdot \vec{\Delta s}$

$\vec{\Delta s} = \vec{r}_f - \vec{r}_i$

$= (5i + 2j + 7k) \cdot [5i - 2j + k - 2i - 3j + 4k]$

$= (5i + 2j + 7k) \cdot (3i - 5j + 5k)$

$= 15 - 10 + 35 =$

$= 40\ J$

Question

A body of mass 4 kg experiences two force $\vec{F}_1 = 5\hat{i} + 8\hat{j} + 7\hat{k}$ and $\vec{F}_2 = 3\hat{i} - 4\hat{j} - 3\hat{k}$. The acceleration acting on the body is: **[JEE Mains 2024]**

1. $-2\hat{i} - \hat{j} - \hat{k}$
2. $4\hat{i} + 2\hat{j} + 2\hat{k}$
3. $2\hat{i} + \hat{j} + \hat{k}$
4. $2\hat{i} + 3\hat{j} + 3\hat{k}$

$$a = \frac{\vec{F_1} + \vec{F_2}}{m}$$

$$a = \frac{8i + 4j + 4k}{4}$$

$$\vec{a} = 2i + j + k$$

Thank

You